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ORIGINAL RESEARCH

CORONARY

Clinical Impact of Diabetes Mellitus After Intravascular Imaging–Guided PCI vs Coronary Artery Bypass Grafting

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ABSTRACT

BACKGROUND Coronary artery bypass grafting (CABG) is the standard revascularization method for patients with diabetes mellitus (DM) and complex coronary artery disease. However, with significant advances in percutaneous coronary intervention (PCI), particularly the use of intravascular imaging (IVI), it is uncertain whether contemporary IVI-guided PCI can achieve clinical outcomes comparable with those of CABG.

OBJECTIVES The aim of this study was to compare the clinical outcomes of IVI-guided PCI, angiography-guided PCI, and CABG in DM patients with left main or 3-vessel disease.

METHODS A total of 3,402 DM patients with left main or 3-vessel disease were included from individual patient-level data of the RENOVATE-COMPLEX-PCI (Randomized Controlled Trial of Intravascular Imaging Guidance Versus Angiography-Guidance on Clinical Outcomes After Complex Percutaneous Coronary Intervention) trial and the institutional registries of Samsung Medical Center (n = 6,962). The primary outcome, which was a composite of all-cause death, nonfatal myocardial infarction, or stroke at 3 years was compared among IVI-guided PCI, angiography-guided PCI, and CABG.

RESULTS In the DM population, the cumulative incidence of the primary outcome at 3 years was 20.1% for angiography-guided PCI, 11.4% for IVI-guided PCI, and 12.4% for CABG. PCI was associated with a significantly higher risk for primary outcome (17.0% vs 12.4%; HR: 1.91; 95% CI: 1.50-2.44; $P < 0.001$). However, the risk for the primary outcome was comparable between the IVI-guided PCI and CABG groups (11.4% vs 12.4%; HR: 0.88; 95% CI: 0.58-1.32; $P = 0.525$). A propensity score-matched analysis yielded similar results (HR: 0.85; 95% CI: 0.53-1.36; $P = 0.507$).

CONCLUSIONS In this hypothesis-generating study, PCI was associated with significantly higher risk for all-cause death, nonfatal myocardial infarction, or stroke at 3 years than CABG in DM patients with left main or 3-vessel disease. However, IVI-guided PCI had comparable risk for clinical events compared with CABG in DM patients. Further randomized controlled trial is needed to confirm this finding. (Randomized Controlled Trial of Intravascular Imaging Guidance Versus Angiography-Guidance on Clinical Outcomes After Complex Percutaneous Coronary Intervention [RENOVATE-COMPLEX-PCI], NCT03381872; institutional cardiovascular catheterization database of Samsung Medical Center [Long-Term Outcomes and Prognostic Factors in Patient Undergoing CABG or PCI], NCT03870815; institutional CABG database of Samsung Medical Center, NCT03870815) (JACC Cardiovasc Interv. 2026;19:555–567) © 2026 by the American College of Cardiology Foundation.

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ABBREVIATIONS AND ACRONYMS

CABG = coronary artery bypass grafting

CAD = coronary artery disease

DES = drug-eluting stent(s)

DM = diabetes mellitus

IVI = intravascular imaging

IVUS = intravascular ultrasound

LM = left main coronary artery

MI = myocardial infarction

PCI = percutaneous coronary intervention

Coronary artery disease (CAD) in patients with diabetes mellitus (DM) is associated with a markedly increased risk for cardiovascular events and mortality. Patients with DM have a 2- to 4-fold higher risk for CAD compared with those without DM, and patients with DM often present with more diffuse and severe coronary atherosclerosis, leading to worse clinical outcomes after coronary revascularization.[1,2] Landmark clinical trials established coronary artery bypass grafting (CABG) as superior to percutaneous coronary intervention (PCI) with first-generation drug-eluting stents (DES) for patients with DM and multivessel CAD.[3] Even the recent FAME 3 (A Comparison of Fractional Flow Reserve-Guided Percutaneous Coronary Intervention and Coronary Artery Bypass Graft Surgery in Patients With Multivessel Coronary Artery Disease) trial failed to show noninferiority of physiology-guided multivessel coronary artery PCI with newer generation DES compared with CABG.[4] These findings have been incorporated into current clinical guidelines, which recommend CABG as the preferred revascularization strategy over PCI in this population.[5,6]

However, PCI has evolved significantly with the use of intravascular imaging (IVI), including intravascular ultrasound (IVUS) and optical coherence tomography for procedural optimization.[7] Recent evidence has now established that IVI-guided PCI significantly improves clinical outcomes compared with angiography-guided PCI, including reduced mortality and target vessel failure.[8-13] This progress creates a critical knowledge gap, as it is unknown whether the benefits of IVI-guided PCI are sufficient to challenge the historical superiority of CABG in the high-risk cohort of patients with DM and complex CAD.

Therefore, the objective of the present study was to compare the long-term clinical outcomes of 3 revascularization strategies—CABG, angiography-guided PCI with contemporary DES, and IVI-guided PCI with contemporary DES—in a pooled analysis from the RENOVATE-COMPLEX-PCI (Randomized Controlled Trial of Intravascular Imaging Guidance Versus Angiography-Guidance on Clinical Outcomes After Complex Percutaneous Coronary Intervention) trial and the institutional PCI and CABG registries of patients with complex CAD, stratified by the presence of DM.

METHODS

STUDY POPULATION AND DATA COLLECTION. This study is a patient-level pooled analysis from 3 large cohorts: the RENOVATE-COMPLEX-PCI trial (NCT03381872), the institutional PCI registry (Long-Term Outcomes and Prognostic Factors in Patient Undergoing CABG or PCI; NCT03870815), and the institutional CABG registry (NCT03870815) from Samsung Medical Center. The RENOVATE-COMPLEX-PCI trial is a prospective, multicenter, randomized controlled trial that enrolled patients with complex CAD from May 2018 to May 2021.[8] The Samsung Medical Center PCI registry included patients undergoing PCI from February 2008 to December 2015, and the CABG registry included patients undergoing isolated CABG between January 2001 and December 2017. Prior studies have been conducted on the basis of these institutional registries.[14,15]

From the combined data set of these 3 cohorts, patients with noncomplex coronary artery lesions, those treated with balloon angioplasty alone, and those treated with stent implantation with either bare-metal stents or first-generation DES were excluded from the institutional PCI registry. The same exclusion criteria were applied to PCI data sets,

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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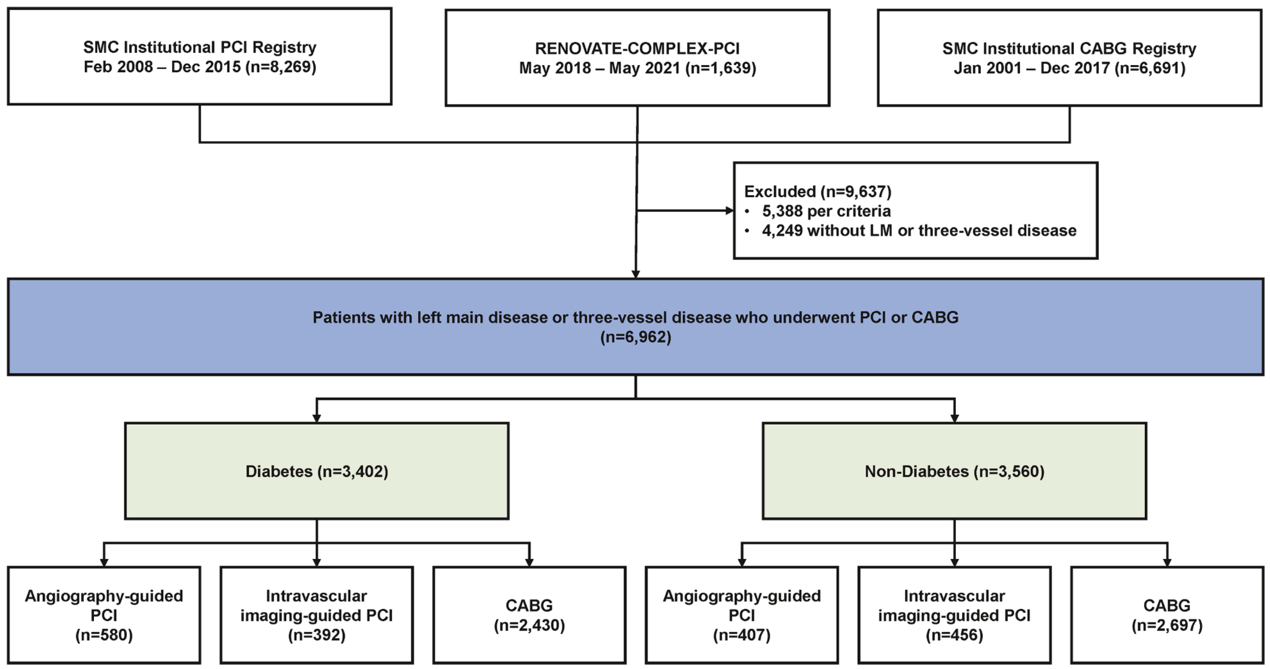
FIGURE 1 **Study Flow**



The present study was patient-level pooled analysis based on 3 large cohorts: RENOVATE-COMPLEX-PCI (Randomized Controlled Trial of Intravascular Imaging Guidance Versus Angiography-Guidance on Clinical Outcomes After Complex Percutaneous Coronary Intervention; NCT03381872), the institutional percutaneous coronary intervention (PCI) registry from Samsung Medical Center (SMC) (NCT03870815), and the institutional coronary artery bypass grafting (CABG) registry from SMC (NCT03870815). In the present analysis, a total of 6,962 patients were divided into 2 groups: 3,402 patients with diabetes mellitus (DM) and 3,560 without DM. The DM group (primary analysis group) was further stratified by revascularization strategy: CABG (n = 2,430), angiography-guided PCI (n = 570), and intravascular imaging (IVI)-guided PCI (n = 392). For comparison, the non-DM group (n = 3,560) had a similar distribution (CABG, n = 2,697; angiography-guided PCI, n = 407; IVI-guided PCI, n = 456).

and those with coronary artery lesions not amenable to perform PCI by operator`s discretion, cardiogenic shock (Killip class IV) at presentation, and known hypersensitivity or contraindication to aspirin, clopidogrel, prasugrel, ticagrelor, heparin, everolimus, or contrast media, or were pregnant or breastfeeding were excluded. From the CABG registry, patients younger than 18 years, those without angiographic data and detailed surgical reports, and those who underwent concomitant valve surgery were excluded. As a result, 6,962 patients with left main coronary artery (LM) or 3-vessel disease who underwent PCI or CABG were included in the present analysis. From this population, 3,402 patients with DM were selected as the primary analysis cohort. Patients were stratified into 3 treatment groups: 1) CABG (n = 2,430); 2) angiography-guided PCI (n = 580); and 3) IVI-guided PCI (n = 392) (Figure 1).

The present study received approval from the Institutional Review Board of Samsung Medical Center and the participating centers. For patients in the institutional registries, the requirement to obtain informed consent was waived by the Institutional Review Board of Samsung Medical Center. Patients from the RENOVATE-COMPLEX-PCI trial provided informed consent prior to enrollment. This study was conducted according to the principles of the Declaration of Helsinki.

TREATMENT. All patients undergoing PCI were treated with second-generation DES.[5,6] In the RENOVATE-COMPLEX-PCI trial, randomization to IVI-guided vs angiography-guided PCI was performed in a 2:1 ratio. The types of IVI devices and the timing of IVI use were at the operator's discretion, but post-PCI evaluation with IVI was mandated for stent optimization under standardized optimization protocols.[8] In the institutional PCI registry, decisions regarding IVI devices and the timing of IVI use were left to the operator's discretion. There were no mandated procedural optimization protocols for IVI in the institutional PCI registry. All patients received guideline-directed medical therapy in accordance with the latest American College of Cardiology/American Heart Association/Society for Cardiovascular Angiography and Interventions or European Society of Cardiology/European Association for Cardio-Thoracic Surgery recommendations, irrespective of whether PCI was performed under IVI or angiographic guidance.[5,6]

CABG was performed according to relevant clinical practice guidelines.[5,6] In the institutional CABG registry, off-pump CABG using bilateral internal thoracic arteries was the standard surgical technique, except in cases in which it was technically unfeasible. The use of cardiopulmonary bypass, the type and number of grafts used, and the anastomosis sites were determined by the heart team discussion and consensus.

CLINICAL OUTCOMES AND DEFINITIONS. The primary outcome of this study was defined as a composite of all-cause death, nonfatal myocardial infarction (MI), or stroke at 3 years. Secondary outcomes were all-cause death, nonfatal MI, stroke, cardiac death, and clinically driven repeat revascularization. Death from unknown causes was defined as cardiac death, and other clinical events were defined according to the definition of the Academic Research Consortium.[16] Stroke was defined as any abrupt neurologic deficit caused by cerebral ischemia or hemorrhage. Nonfatal MI was defined using the third universal definition of MI, while procedure-related MIs were excluded from the analysis.[17]

Patient follow-up was performed through outpatient clinic visits scheduled at 1, 6, and 12 months and annually thereafter. Patients who were unable to attend these appointments were contacted via telephone. Data for the institutional registries were gathered prospectively by research coordinators and augmented with information from medical records or phone interviews where required. Survival status for the entire cohort was verified against the Korean National Health Insurance database. The follow-up period for the analysis was truncated at 3 years postprocedure, at which point all data were censored.

STATISTICAL ANALYSIS. The normality of the continuous variables was evaluated using the Kolmogorov-Smirnov test. Data are presented as mean ± SD for normally distributed variables and median (Q1-Q3) for non-normally distributed variables. Group comparisons were performed using Welch's *t*-test for normally distributed variables and the Mann-Whitney *U* test for non-normally distributed variables.

Categorical variables are expressed as numbers and percentages and were compared using the chi-square test. To analyze clinical outcomes over time, we estimated the 3-year cumulative incidence using the Kaplan-Meier method and assessed overall differences between survival curves with the log-rank test. HRs and 95% CIs for the entire follow-up period were calculated using Cox proportional hazards models. The proportional hazards assumption was verified using a 2-sided score test of the scaled Schoenfeld residuals ($P < 0.05$). To adjust for significant baseline differences between the nonrandomized treatment groups, 2 distinct statistical risk adjustment methods were used. To adjust for potential confounding variables in the overall cohort analysis, multivariable Cox proportional hazards regression and propensity score matching analysis were used. Propensity scores were calculated for each patient using a full nonparsimonious logistic regression model with the treatment group as the dependent variables (Supplemental Table 1). Propensity score matching was conducted using the nearest neighbor method with a 0.2-SD caliper width. This was performed for 3 distinct comparisons: CABG vs PCI, CABG vs IVI-guided PCI, and CABG vs angiography-guided PCI. A 2:1 matching ratio was specifically used for the IVI-guided PCI comparison to accommodate its smaller group size. Rules for matching included random selection for multiple matches and exclusion of any unmatched patients. The postmatching balance of covariates was evaluated by assessing the absolute standardized mean differences. In addition, inverse probability treatment weighting analysis was performed on the basis of propensity scores. All statistical tests were 2-sided, and P values <0.05 were considered to indicate statistical significance. All analyses were performed using R version 4.2.1 (R Foundation for Statistical Computing).

RESULTS

BASELINE DEMOGRAPHIC, ANGIOGRAPHIC, AND PROCEDURAL CHARACTERISTICS. In the DM population, patients who underwent CABG were younger than those in the PCI groups but had a higher prevalence of previous stroke and peripheral vascular disease. The prevalence of dyslipidemia was highest in the IVI-guided PCI group, while current smoking was most prevalent in the CABG group (**Table 1**, Supplemental Table 2). Three-vessel disease was most common in the CABG group in both the DM and non-DM populations, while LM disease was more frequent in the IVI-guided PCI groups. The mean

TABLE 1 Baseline Clinical Characteristics According to Treatment Group in Patients With Left Main Disease or 3-Vessel Disease With Diabetes

	Angiography (n = 580)	IVI (n = 392)	CABG (n = 2,430)	P Value[a]	P Value[b]	P Value[c]
Demographic characteristics						
Age, y	67.4 ± 10.4	65.4 ± 9.9	64.4 ± 8.8	0.057	<0.001	<0.001
Male	411 (70.9)	308 (78.6)	1,740 (71.6)	0.005	0.761	0.012
Body mass index, kg/m^2	24.5 ± 3.2	24.5 ± 3.0	24.5 ± 3.1	0.839	0.980	0.980
Cardiovascular risk factors						
Hypertension	427 (73.6)	278 (70.9)	1,713 (70.5)	0.911	0.149	0.328
Dyslipidemia	202 (34.8)	179 (45.7)	768 (31.6)	<0.001	0.149	<0.001
Current smoking	113 (19.5)	66 (16.8)	746 (30.7)	<0.001	<0.001	<0.001
Chronic kidney disease	88 (15.2)	36 (9.2)	279 (11.5)	0.210	0.018	0.010
Previous PCI	119 (20.5)	90 (23.0)	461 (19.0)	0.075	0.430	0.158
Previous CABG	47 (8.1)	14 (3.6)	31 (1.3)	0.002	<0.001	<0.001
Previous peripheral vascular disease	28 (4.8)	18 (4.6)	231 (9.5)	0.002	<0.001	<0.001
Previous stroke	61 (10.5)	33 (8.4)	453 (18.6)	<0.001	<0.001	<0.001
Clinical manifestation						
Initial presentation				<0.001	<0.001	<0.001
Chronic coronary syndrome	312 (53.8)	227 (57.9)	1,042 (42.9)			
Acute coronary syndrome	268 (46.2)	165 (42.1)	1,388 (57.1)			
Acute myocardial infarction	151 (26.0)	59 (15.1)	500 (20.6)	0.013	0.005	<0.001
Left ventricular ejection fraction, %	55.7 ± 12.2	56.6 ± 12.1	52.4 ± 14.3	<0.001	<0.001	<0.001
Mechanical circulatory support	27 (4.7)	4 (1.0)	79 (3.3)	0.024	0.128	0.007
Laboratory results						
Hemoglobin	12.8 ± 2.2	13.2 ± 2.5	12.6 ± 1.8	<0.001	0.024	<0.001
Creatinine	1.5 ± 1.6	1.3 ± 1.7	1.4 ± 1.5	0.418	0.376	0.407
Peak CK-MB	5.4 (2.7-9.4)	3.5 (2.1-11.1)	1.4 (0.8-2.9)	<0.001	<0.001	<0.001
Peak troponin I or troponin T	0.3 (0.1-2.3)	0.2 (0.0-0.8)	0.1 (0.0-0.4)	<0.001	<0.001	<0.001
NT-proBNP	846.3 (186.7-2,790.0)	249.9 (63.1-908.3)	283.5 (96.3-1,011.5)	0.621	<0.001	<0.001
Medications at discharge						
Aspirin	538 (92.8)	372 (94.9)	2,396 (98.6)	<0.001	<0.001	<0.001
P2Y$_{12}$ inhibitor	558 (96.2)	367 (93.6)	1,424 (58.6)	<0.001	<0.001	<0.001
Statin	535 (92.2)	356 (90.8)	1,860 (76.5)	<0.001	<0.001	<0.001
Beta-blocker	306 (52.8)	193 (49.2)	1,788 (73.6)	<0.001	<0.001	<0.001
Renin-angiotensin system blocker	298 (51.4)	222 (56.6)	900 (37.0)	<0.001	<0.001	<0.001

Values are mean ± SD, n (%), or median (Q1-Q3). [a]*P* value from comparison between IVI-guided PCI and CABG groups. [b]*P* value from comparison between angiography-guided PCI and CABG groups. [c]*P* value from comparison among the 3 groups (angiography-guided PCI, IVI-guided PCI, and CABG).

CABG = coronary artery bypass grafting; CK = creatine kinase; IVI = intravascular imaging; NT-proBNP = N-terminal pro-brain natriuretic peptide; PCI = percutaneous coronary intervention.

number of treated lesions per patient was greatest in the CABG group across both the DM and non-DM populations (**Table 2**, Supplemental Table 2).

CLINICAL OUTCOMES BETWEEN PCI AND CABG. The median follow-up duration was 3.1 years (Q1-Q3: 2.4-4.5 years). At 3 years, the cumulative incidence of the primary outcome was significantly higher in the PCI group than in the CABG group among patients with DM (17.0% vs 12.4%; adjusted HR: 1.91; 95% CI: 1.50-2.44; $P < 0.001$) (**Figure 2**). A significantly higher risk for the primary outcome was driven by a higher incidence of all-cause death, cardiac death, nonfatal MI, and repeat revascularization (Supplemental Table 3). In the non-DM population, there was no significant difference in the cumulative incidence of primary outcome between the PCI group and the CABG groups (9.0% vs 9.4%; adjusted HR: 1.22; 95% CI: 0.86-1.75; $P = 0.268$) (**Figure 2**). However, the PCI group showed a significantly higher risk for cardiac death, nonfatal MI, and repeat revascularization (Supplemental Table 3).

CLINICAL OUTCOMES AMONG ANGIOGRAPHY-GUIDED PCI, IVI-GUIDED PCI, AND CABG. When PCI was classified according to angiography-guided PCI or IVI-guided PCI, the incidence of primary outcome was highest in the angiography-guided PCI group in both the DM and non-DM populations, which was significantly higher than the CABG group. Furthermore, angiography-guided PCI showed significantly higher risk for secondary outcomes than the IVI-guided PCI or CABG groups in both the DM and non-DM populations (**Table 3**, Supplemental Table 4).

Conversely, the cumulative incidence of the primary outcome was comparable between the IVI-guided PCI and CABG groups in the DM population (11.4% vs 12.4%; adjusted HR: 0.88; 95% CI:

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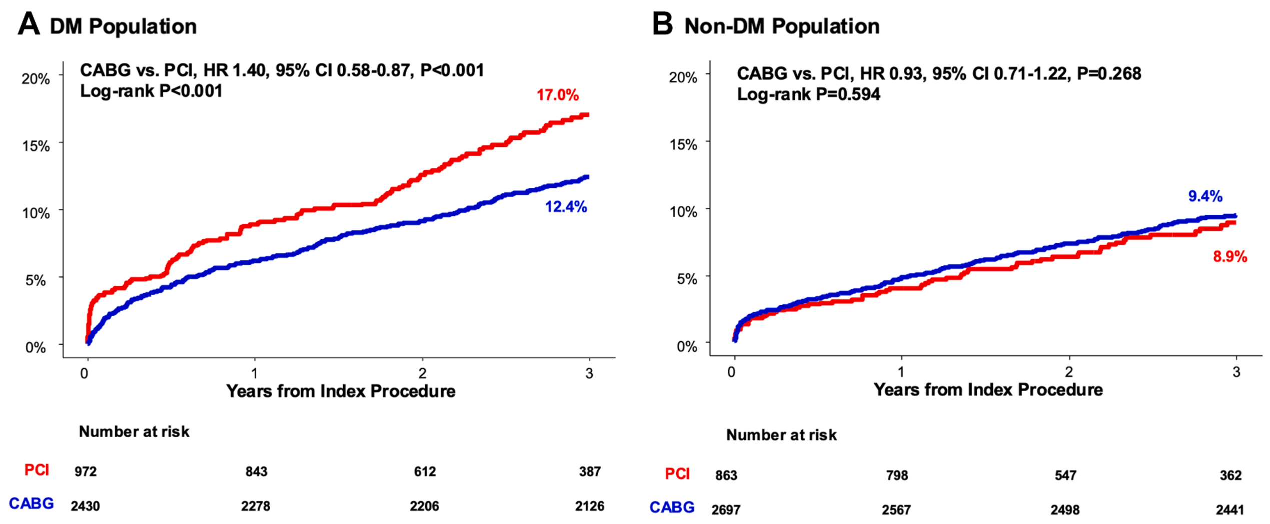
TABLE 2 Baseline Procedural Characteristics According to Treatment Group in Patients With Left Main Disease or 3-Vessel Disease With Diabetes

	Angiography (n = 580)	IVI (n = 392)	CABG (n = 2,430)	*P* Value[a]	*P* Value[b]	*P* Value[c]
Procedure duration, min	82.4 ± 55.8	91.5 ± 50.1	356.4 ± 87.1	<0.001	<0.001	<0.001
Contrast volume, mL	223.4 ± 85.9	229.1 ± 98.2	NA	NA	NA	NA
Transradial approach	409 (70.5)	271 (69.1)	NA	NA	NA	NA
Angiographic disease severity						
3-vessel disease	517 (89.1)	249 (63.5)	2,269 (93.4)	<0.001	0.001	<0.001
Left main coronary artery	143 (24.7)	219 (55.9)	543 (22.3)	<0.001	0.256	<0.001
Number of complex coronary lesions	2.6 ± 1.2	3.0 ± 1.4	NA	NA	NA	NA
Bifurcation	196 (33.8)	196 (50.0)	NA	NA	NA	NA
Chronic total occlusion	121 (20.9)	60 (15.3)	NA	NA	NA	NA
Long lesion (≥38 mm)	329 (56.7)	225 (57.4)	NA	NA	NA	NA
Multivessel disease (≥2 epicardial vessels)	437 (75.3)	235 (59.9)	NA	NA	NA	NA
Multiple stents (≥3 stents per patient)	176 (30.3)	110 (28.1)	NA	NA	NA	NA
In-stent restenosis	39 (6.7)	36 (9.2)	NA	NA	NA	NA
Severely calcified lesions	37 (6.4)	47 (12.0)	NA	NA	NA	NA
Ostial lesions	93 (16.0)	97 (24.7)	NA	NA	NA	NA
Number of treated lesions	3.9 ± 1.6	2.6 ± 1.4	4.3 ± 1.1	<0.001	<0.001	<0.001
Mean diameter stenosis before PCI	85.9 ± 8.9	84.0 ± 10.8	NA	NA	NA	NA
Mean diameter stenosis after PCI	6.1 ± 10.6	6.6 ± 10.3	NA	NA	NA	NA
Number of stents per patient	2.2 ± 1.1	2.1 ± 1.1	NA	NA	NA	NA
Mean stent diameter, mm	2.9 ± 0.4	3.3 ± 1.9	NA	NA	NA	NA
Total stent length, mm	56.0 ± 31.6	56.0 ± 34.6	NA	NA	NA	NA
Type of stent				NA	NA	NA
Second-generation drug-eluting stent	557 (96.0)	382 (97.4)	NA	NA	NA	NA
Drug-coated balloon	23 (4.0)	10 (2.6)	NA	NA	NA	NA
Adjunctive balloon dilatation	183 (31.6)	254 (64.8)	NA	NA	NA	NA
Rotablation	21 (3.6)	27 (6.9)	NA	NA	NA	NA
Off-pump CABG	NA	NA	1,928 (79.3)	NA	NA	NA
Pump time, min	NA	NA	118.0 ± 55.6	NA	NA	NA
Aortic cross-clamp time, min	NA	NA	97.5 ± 34.8	NA	NA	NA
Number of conduits per patient	NA	NA	2.3 ± 0.5	NA	NA	NA
Left internal thoracic artery conduit	NA	NA	2,380 (97.9)	NA	NA	NA
Right internal thoracic artery conduit	NA	NA	2,288 (94.2)	NA	NA	NA
Right gastroepiploic artery conduit	NA	NA	339 (14.0)	NA	NA	NA
Radial artery conduit	NA	NA	52 (2.1)	NA	NA	NA
Saphenous vein graft conduit	NA	NA	568 (23.4)	NA	NA	NA
Anastomosis sites						
Left anterior descending coronary artery	NA	NA	2,398 (98.7)	NA	NA	NA
D1	NA	NA	1,572 (64.7)	NA	NA	NA
D2	NA	NA	199 (8.2)	NA	NA	NA
D3	NA	NA	15 (0.6)	NA	NA	NA
Ramus intermedius	NA	NA	362 (14.9)	NA	NA	NA
OM1	NA	NA	2,101 (86.5)	NA	NA	NA
OM2	NA	NA	833 (34.3)	NA	NA	NA
OM3	NA	NA	213 (8.8)	NA	NA	NA
Right coronary artery	NA	NA	276 (11.4)	NA	NA	NA
Posterior descending coronary artery	NA	NA	1,679 (69.1)	NA	NA	NA
Posterolateral branch	NA	NA	690 (28.4)	NA	NA	NA
Hospitalization, d	2.0 (1.0-5.0)	2.0 (1.0-3.0)	10.0 (7.0-14.0)	<0.001	<0.001	<0.001

Values are mean ± SD, n (%), or median (Q1-Q3). [a]*P* value from comparison between IVI-guided PCI and CABG groups. [b]*P* value from comparison between angiography-guided PCI and CABG groups. [c]*P* value from comparison among the 3 groups (angiography-guided PCI, IVI-guided PCI, and CABG).

D = diagonal artery; NA = not applicable; OM = obtuse marginal artery; other abbreviations as in Table 1.

FIGURE 2 **Comparison of 3-Year Clinical Outcomes Between PCI and CABG in Patients With Left Main or 3-Vessel Disease, Stratified by DM Status**



Kaplan-Meier curves for the primary composite outcome (all-cause death, nonfatal myocardial infarction, or stroke) at 3 years comparing PCI vs CABG in patients with (A) and those without (B) DM. Numbers at risk are shown; comparisons were made using log-rank and Cox models. Abbreviations as in Figure 1.

TABLE 3 **Primary and Secondary Outcomes According to Treatment Groups in Diabetes Mellitus Patients With Left Main Disease or 3-Vessel Disease**

				Total Population				Propensity Score-Matched Population[a]			
				IVI vs CABG		Angiography vs CABG		IVI vs CABG		Angiography vs CABG	
	Angiography (n = 580)	IVI (n = 392)	CABG (n = 2,430)	Adjusted HR[b] (95% CI)	*P* Value	Adjusted HR[b] (95% CI)	*P* Value	HR (95% CI)	*P* Value	HR (95% CI)	*P* Value
Primary outcome[c]	104 (20.6)	36 (11.4)	302 (12.4)	0.88 (0.58-1.32)	0.525	1.88 (1.47-2.40)	<0.001	0.85 (0.53-1.36)	0.507	2.01 (1.46-2.77)	<0.001
All-cause death	74 (14.7)	20 (5.8)	265 (10.9)	0.59 (0.36-0.99)	0.045	1.47 (1.11-1.94)	0.008	0.57 (0.31-1.03)	0.062	1.70 (1.18-2.44)	0.004
Cardiac death	52 (10.3)	11 (3.0)	60 (2.5)	1.27 (0.58-2.79)	0.555	4.61 (3.03-7.01)	<0.001	1.12 (0.49-2.56)	0.795	3.86 (2.16-6.88)	<0.001
Nonfatal MI	41 (8.6)	12 (4.2)	14 (0.6)	7.38 (2.59-21.1)	<0.001	16.3 (8.44-31.4)	<0.001	3.47 (1.25-9.62)	0.017	15.3 (4.73-49.8)	<0.001
Stroke	22 (4.4)	6 (2.2)	41 (1.8)	0.44 (0.16-1.25)	0.123	2.08 (1.17-3.71)	0.013	0.47 (0.13-1.67)	0.243	1.82 (0.91-3.64)	0.092
Repeat revascularization	72 (14.9)	43 (12.8)	48 (2.1)	7.37 (4.30-12.6)	<0.001	7.82 (0.27-2.80)	<0.001	5.00 (2.73-9.13)	<0.001	6.77 (3.65-12.5)	<0.001

Values are n (%). [a]Adjusted variables by propensity score matching analysis between PCI and CABG groups were age, male sex, dyslipidemia, current smoking, previous CABG, previous peripheral vessel disease, previous stroke, acute coronary syndrome, left ventricular ejection fraction, 3-vessel disease, left main coronary artery disease, number of treated lesions per patient, and mechanical circulatory support. Between CABG and IVI-guided PCI, matching analysis was performed with a 2:1 ratio. Between CABG and angiography-guided PCI, matching analysis was performed with a 1:1 ratio. [b]Adjusted variables by multivariate analysis according to treatment groups were age, male sex, dyslipidemia, current smoking, previous CABG, previous peripheral vessel disease, previous stroke, acute coronary syndrome, left ventricular ejection fraction, 3-vessel disease, left main coronary artery disease, number of treated lesions per patient, and mechanical circulatory support. [c]The primary outcome is a composite of all-cause death, nonfatal MI, or stroke.

MI = myocardial infarction; other abbreviations as in Table 1.

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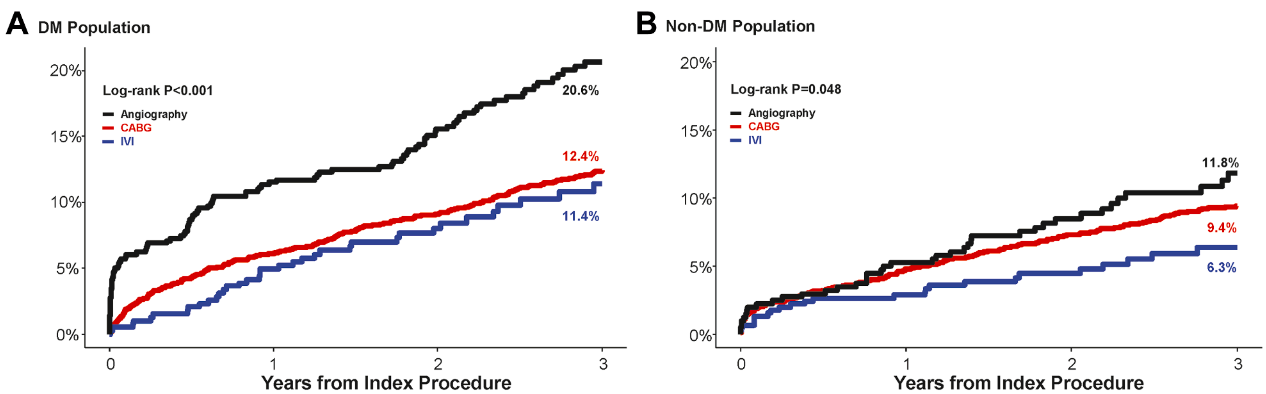
FIGURE 3 **Comparison of 3-Year MACCE Among Patients With Left Main or 3-Vessel Disease According to Revascularization Strategy, Stratified by DM Status**



A DM Population — Number at risk

	0	1	2	3
Angiography	580	484	369	242
CABG	2430	2278	2206	2126
IVI	392	359	243	142

	Hazard ratio	95% CI	P value
CABG	Reference	Reference	Reference
Angiography	1.77	1.41-2.20	<0.001
IVI	0.88	0.63-1.25	0.482

B Non-DM Population — Number at risk

	0	1	2	3
Angiography	407	367	265	174
CABG	2697	2567	2498	2441
IVI	456	431	282	188

	Hazard ratio	95% CI	P value
CABG	Reference	Reference	Reference
Angiography	1.23	0.88-1.72	0.220
IVI	0.66	0.43-1.00	0.050

Kaplan-Meier curves for the primary composite outcome (all-cause death, nonfatal myocardial infarction, or stroke) at 3 years comparing IVI-guided PCI, angiography-guided PCI, and CABG in patients with (A) and those without (B) DM. Numbers at risk are shown; comparisons were made using log-rank and Cox models. Abbreviations as in Figure 1.

0.58-1.32; $P = 0.525$) as well as in the non-DM population (6.3% vs 9.4%; adjusted HR: 0.77; 95% CI: 0.48-1.22; $P = 0.260$) (Figure 3, Table 3). In both the DM and non-DM populations, IVI-guided PCI showed a lower risk for all-cause death than the CABG group, but it was followed by higher risk for nonfatal MI and repeat revascularization (Table 3, Supplemental Table 4, Supplemental Figure 1). Propensity score-matched analysis and inverse probability treatment weighting analysis showed similar findings (Table 3, Supplemental Table 4, Supplemental Table 5). These results were similarly observed when LM and 3-vessel disease were separately evaluated (Supplemental Figure 2).

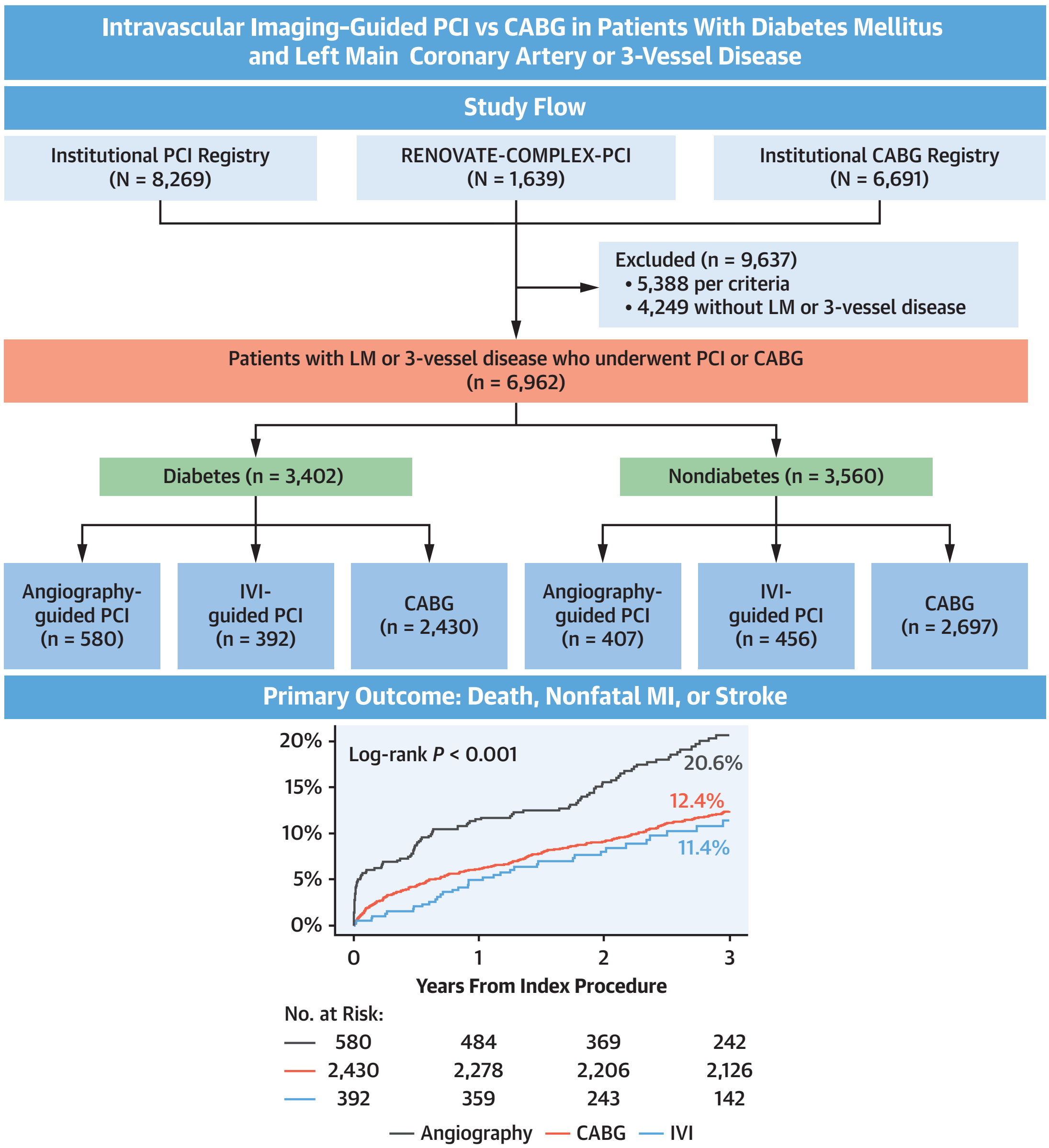
DISCUSSION

The aim of the present hypothesis-generating analysis, based on a pooled individual patient-level data set from the RENOVATE-COMPLEX-PCI trial and 2 large institutional registries, was to compare the long-term clinical outcomes of IVI-guided PCI vs CABG in patients with LM or 3-vessel CAD. The main findings were as follows.

First, among patients with DM, PCI was associated with a significantly higher risk for the primary outcome, which was a composite of all-cause death, nonfatal MI, or stroke, compared with CABG. Second, angiography-guided PCI consistently showed significantly worse clinical outcomes than IVI-guided PCI or CABG. Third, unlike angiography-guided PCI, IVI-guided PCI showed a comparable risk for primary outcome compared with CABG (Central Illustration).

Patients with DM typically have more complex and diffuse coronary atherosclerotic disease than those without DM, characterized by heightened inflammatory responses within vessels, progressive atherosclerosis, a higher incidence of restenosis, and extensive lesion distribution.[18,19] CABG offers an advantage in this population, providing myocardial perfusion regardless of lesion complexity or anatomical constraints, and potential protection against vessel occlusion from non-flow-limiting stenosis in grafted vessels.[20] In contrast, PCI is local

CENTRAL ILLUSTRATION **Study Overview and Main Findings**



Among patients with diabetes mellitus and LM or 3-vessel disease:

- PCI was associated with a higher 3-year risk of death, nonfatal MI, or stroke as compared with CABG.
- However, IVI-guided PCI showed rates of death, nonfatal MI, or stroke comparable with CABG.

Gim DH, et al. JACC Cardiovasc Interv. 2026;19(5):555-567.

Comparison of intravascular imaging (IVI)-guided percutaneous coronary intervention (PCI) and coronary artery bypass grafting (CABG) in patients with diabetes mellitus and left main coronary artery (LM) or 3-vessel disease. In a pooled analysis of 3 large registries, overall PCI showed a higher 3-year risk for death, nonfatal myocardial infarction (MI), or stroke than CABG. However, IVI-guided PCI showed comparable risk for death, nonfatal MI, or stroke compared with CABG. RENOVATE-COMPLEX-PCI = Randomized Controlled Trial of Intravascular Imaging Guidance Versus Angiography-Guidance on Clinical Outcomes After Complex Percutaneous Coronary Intervention.

treatment rendering complete revascularization, which is particularly challenging in diabetic patients who present complex multivessel disease. This limitation increases the risk for restenosis and repeat revascularization in diabetic patients undergoing PCI.[21] Indeed, CABG has consistently demonstrated superior clinical outcomes compared with PCI in diabetic patients with multivessel CAD.[3,22] Even with the use of newer generation DES and physiology-guided selective treatment of multivessel CAD, the recent FAME 3 trial failed to show noninferiority of PCI compared with CABG in both DM and non-DM patients.[4,23] On the basis of these results, current guidelines strongly recommend CABG as the preferred revascularization strategy in diabetic patients with multivessel disease.[5,24]

However, previous research has not addressed the potential benefits of IVI-guided PCI compared with CABG. Recent trials have consistently shown superior clinical outcomes following IVI-guided PCI over angiography-guided PCI,[8-13] and the prognostic benefit of IVI-guided PCI was more prominent in patients with complex CAD, including LM or 3-vessel disease.[13,25] In the subgroup analysis of diabetic patients from the RENOVATE-COMPLEX-PCI trial, IVI-guided PCI was associated with a similar reduction in target vessel failure compared with angiography-guided PCI, and the interaction between DM status and treatment effect was not statistically significant.[8] Similarly, in the DM subgroup of the ULTIMATE (Intravascular Ultrasound Guided Drug Eluting Stents Implantation in “All-Comers” Coronary Lesions) trial, IVUS-guided PCI significantly reduced the risk for target vessel failure compared with angiography-guided PCI, with no significant interaction by DM.[26] Most recently, the IVUS-ACS trial demonstrated that in diabetic patients with acute coronary syndrome, IVUS-guided PCI was associated with a significantly lower rate of target vessel failure compared with angiography-guided PCI.[27] Despite accumulating evidence supporting IVI-guided PCI, direct comparative studies between IVI-guided PCI and CABG are limited. A recent study compared IVI-guided PCI with CABG in patients with multivessel disease, reporting no significant difference in clinical outcomes, consistent with our findings.[28] However, large-scale studies or randomized controlled trials directly comparing IVI-guided PCI with CABG specifically in diabetic patients remain limited.

The present study demonstrated that in diabetic patients, PCI was associated with a significantly higher risk for the primary composite outcome (death, nonfatal MI, or stroke) at 3 years compared with CABG. The significantly worse clinical outcome in the PCI group was driven mainly by a higher risk for death, nonfatal MI, and stroke in the angiography-guided PCI group. Conversely, IVI-guided PCI showed a comparable risk for the primary outcome as with CABG, and propensity score-matched analysis showed similar results. Nonetheless, IVI-guided PCI still showed higher rates of nonfatal MI and repeat revascularization compared with CABG. This might indicate intrinsic limitations of PCI. A previous landmark trial, the FREEDOM (Comparison of Two Treatments for Multivessel Coronary Artery Disease in Individuals With Diabetes) trial, demonstrated the superiority of CABG over PCI in patients with DM and multivessel CAD, leading to the Class 1 recommendation for CABG in current guidelines.[3,6] However, that trial was conducted before the widespread use of contemporary DES and did not clearly specify the use of IVUS or optical coherence tomography. Considering that IVI-guided PCI has been shown to be superior to angiography-guided PCI,[8,9,12,29] these findings raise an important clinical question as to whether IVI-guided PCI could achieve outcomes comparable with those of CABG in this high-risk population. In this context, the present study provides additional evidence by directly comparing angiography-guided PCI, IVI-guided PCI, and CABG among patients with DM, which is an analysis that has not been performed in randomized controlled trials. The present results showing a comparable risk for composite outcomes between IVI-guided PCI and CABG highlight the importance of procedural optimization guided by IVI in patients with LM or 3-vessel disease.[30] It is well established that IVI enables more precise stent landing through detailed assessment of plaque burden and stent-edge morphology and facilitates optimal stent expansion and apposition. These advantages may partly explain the comparable composite risk for death, nonfatal MI, or stroke between IVI-guided PCI and CABG in diabetic patients with complex coronary anatomy. Moving forward, the research paradigm should shift from a generalized comparisons between CABG and PCI to specific comparisons between CABG and IVI-guided PCI. Therefore, additional large-scale studies and randomized trials are crucial to clarify comparative prognosis between IVI-guided PCI and CABG in this high-risk population.

STUDY LIMITATIONS. First, the nonrandomized, observational nature of the present analysis inherently carries potential biases, including selection bias and unmeasured confounding factors. Although the included registries were prospective registries, selection or information bias could affect the independent variables compared with randomized controlled trials. Unmeasured confounding factors such as frailty or functional capacity might have influenced the treatment decision, which could not be fully accounted for even after adjustment by multivariable modes and propensity score matching analysis.

Second, the inclusion period varied across the different registries and the trial, introducing potential temporal bias due to evolving treatment standards and technologies overtime. Third, the CABG group was derived from a single-center registry from one of the largest tertiary hospitals in Korea, which may also be susceptible to bias and limit the generalizability of the results.

Fourth, although previous trials comparing PCI and CABG have presented angiographic index such as SYNTAX (Synergy Between PCI With Taxus and Cardiac Surgery) score to evaluate anatomical complexity, the data set of the present study did not systematically collect these data.

Fifth, we analyzed outcomes according to the presence or absence of DM, without incorporating analyses according to the severity of DM. Furthermore, lipid and glycemic target attainment could not be evaluated, as the study neither mandated treatment targets nor uniformly collected serial low-density lipoprotein cholesterol and glycated hemoglobin values. However, patients received medical therapy according to current clinical guidelines.

Sixth, the results of this study are limited to outcomes within a 3-year follow-up period and cannot provide information on longer term results. Extended follow-up to 5 years is ongoing across the contributing cohorts.

Seventh, although our study provides robust hypothesis-generating findings, the lack of randomization specifically between IVI-guided PCI and CABG restricts definitive conclusion on causality. Finally, these data stem largely from a homogenous population of Korean patients, so the results might not be generalizable to other patient populations.

CONCLUSIONS

In this hypothesis-generating study of patients with DM and LM or 3-vessel CAD, PCI was associated with a significantly higher risk for a composite outcome of death, nonfatal MI, or stroke compared with CABG. However, IVI-guided PCI demonstrated a comparable risk for clinical events relative to CABG. Further randomized controlled trials are warranted to confirm these observations and establish definitive treatment strategy.

DATA AVAILABILITY STATEMENT Anonymized patient-level data will be made available by the corresponding author for reasonable requests. Consent was not obtained for data sharing, but the presented data are anonymized, and the risk for identification is minimal.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

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PERSPECTIVES

WHAT IS KNOWN? CABG has been the preferred strategy for patients with DM and complex CAD. Although IVI-guided PCI significantly improved clinical outcomes compared with angiography-guided PCI, comparative evidence between IVI-guided PCI and CABG remains limited.

WHAT IS NEW? In this pooled analysis, IVI-guided PCI showed comparable risk for a composite of death, nonfatal MI, and stroke at 3 years with CABG, whereas angiography-guided PCI showed significantly higher risk for clinical events compared with CABG.

WHAT IS NEXT? A randomized trial in patients with DM comparing IVI-guided PCI and CABG, incorporating extended follow-up and patient-centered outcomes, is needed to confirm the present observations.

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KEY WORDS coronary artery bypass grafting, coronary artery disease, diabetes mellitus, intravascular ultrasound, percutaneous coronary intervention

APPENDIX For supplemental figures and tables, and a video of the interactive Central Illustration, please see the online version of this paper.